**Tertiary Infotech Academy Pte Ltd**

UEN: 201200696W

# LESSON PLAN

For

## Agentic AI Applications with Claude Code

TGS Ref No: TGS-2025052468

Conducted by

**Tertiary Infotech Academy Pte Ltd**

UEN: 201200696W

**Version 2.5**



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## DOCUMENT VERSION CONTROL RECORD

| Version Number | Effective Date of Release | Summary of Included Changes | Author |
|---|---|---|---|
| 1.0 | 27 Jun 2026 | First version — full-day (9:00 AM–6:00 PM) lesson plan mapped to the three topics and activities | Dr. Alfred Ang |
| 1.1 | 27 Jun 2026 | One-day schedule with 1-hour assessment at 5:00 PM (30 min WA + 30 min PP), shortened breaks, and courseware/assessment on the LMS | Tertiary Infotech Academy Pte Ltd |
| 1.2 | 27 Jun 2026 | Aligned Topic 1 to the 7-step build workflow (Activity 1a/1b) and added the Hooks activity | Tertiary Infotech Academy Pte Ltd |
| 1.3 | 8 Jul 2026 | Added a "Setup Claude Code in VS Code" slide (now slide 19) and realigned all slide-number references to the updated 65-slide deck (v13.0) | Tertiary Infotech Academy Pte Ltd |
| 1.4 | 2 Sep 2026 | Added Claude for Chrome & Claude for Science to the Claude Product Family slide; added a "Plan" step to the Agentic Loop in Action slide; added a new "SKILL.md vs CLAUDE.md" slide. Slide references updated. | Tertiary Infotech Academy Pte Ltd |
| 1.5 | 2 Sep 2026 | Added a Practice Exam Portal slide (exams.tertiaryinfotech.com) and an LMS screenshot on the Courseware & Assessment slide; reworked Activity 6 to the floating WhatsApp widget with a 10-second auto-open hook, suggested queries and voice activation. | Tertiary Infotech Academy Pte Ltd |
| 1.6 | 4 Sep 2026 | Added a "User Level | Tertiary Infotech |


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| | | vs Project Level" slide (slide 42) comparing where commands, MCP tools, skills, CLAUDE.md and hooks are stored at user level (applies to all projects) versus project level (one project only); realigned all slide-number references to the 68-slide deck (v14.0) | Academy Pte Ltd |
| 1.7 | 9 Sep 2026 | Added a "What Triggers What" key-concepts summary slide (new slide 64) covering CLAUDE.md, skills, tools, commands, hooks and sub agents and their triggers; expanded the SKILL.md vs CLAUDE.md comparison. | Tertiary Infotech Academy Pte Ltd |
| 1.8 | 9 Sep 2026 | Corrected Topic 1.2 / Activity 1 slide references (31–38 and 25–30, 39). Added the Lab Materials section covering the four restructured lab folders (Lab 1, Lab 1b, Lab 2, Lab 3). | Tertiary Infotech Academy Pte Ltd |
| 1.9 | 9 Sep 2026 | Recomputed every slide reference against deck v15 after the Lab Materials slide insert shifted numbering by one: Topic 1.2 = 32–39, Activity 1 = 31 and 40, and all other schedule rows and topic headings realigned. | Tertiary Infotech Academy Pte Ltd |
| 2.0 | 29 Oct 2026 | Recomputed every slide reference against deck v21 (77 slides) after seven setup/GitHub | Tertiary Infotech Academy Pte Ltd |


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| | | screenshot slides were added and the agentic-loop slides moved to follow Harness Engineering (slides 19–22) and the setup walkthrough moved into Activity 1 after the 7 Steps (29–31): Topic 1 = 14–49 (Activity 1 = 28–32, 37, 46–49), Topic 2 = 50–59, Topic 3 = 60–73, closing = 72–77. Replaced the retired Mini Capstone with Activities 6–7 to match the deck. | |
| 2.1 | 29 Oct 2026 | Recomputed slide references against deck v22 (79 slides) after two hook-example slides were added after Automating with Hooks (slides 70–71): Topic 3 = 60–75, Topic 3.2/3.3 = 64–71, Activities 6–7 = 72–73, closing = 74–79. | Tertiary Infotech Academy Pte Ltd |
| 2.2 | 31 Oct 2026 | Slide references recomputed for deck v23 (83 slides; new slides 19–20, 64, 68): Topic 1 = 14–51, Topic 2 = 52–61, Topic 3 = 62–79, closing = 78–83. Added the seven parts of an AI agent and skill vs sub agent front matter. | Tertiary Infotech Academy Pte Ltd |
| 2.3 | 7 Nov 2026 | Restructured to one topic, one activity, one lab and one PP task, aligned with deck v25 (91 slides). Setup now covers four surfaces, including the cloud at claude.ai/code, with new slides for saved | Tertiary Infotech Academy Pte Ltd |


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| | | Desktop environments and cloud sessions (34–35). Agent Skills moved to Topic 2, now Commands, Tools and Skills (Activity 2A–2D; 2D is the evidence for PP Task 2). Topic 3 is now Agents, Hooks and Security (Activity 3A–3C): learners build their own hooks (PP Task 3), with a new PreToolUse/PostToolUse use-case slide and five new AI security slides, one per principle. The WhatsApp widget activity is removed. Afternoon re-timed within the same 1:15–5:00 PM block; slide references recomputed: Topic 1 = 14–53, Topic 2 = 54–67, Topic 3 = 68–87, closing = 87–91. Labs repository link corrected. | |
| 2.4 | 8 Nov 2026 | Added a RAG chatbot + hooks slide (new slide 77): the bot retrieves the refund policy, a PreToolUse hook validates the refund request and a PostToolUse hook processes the refund. Slide references recomputed for deck v26 (92 slides): Topic 3 = 68–88, Hooks = 74–80, AI Security = 81–87, closing = 88–92. | Tertiary Infotech Academy Pte Ltd |
| 2.5 | 13 Nov 2026 | Aligned with deck v27 (95 slides). New slides: /usage after Claude Code Pricing | Tertiary Infotech Academy Pte Ltd |


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| | | (slide 30), the Claude Desktop permission-mode menu after Permission Modes (slide 39) and /context after The Context Window (slide 48). Slide 42 now shows the current Desktop Settings → Code page, and the trainer slide adds the Claude Certified Architect – Foundations (CCAR-F) badge. Slide references recomputed: Topic 1 = 14–56, Topic 2 = 57–70, Topic 3 = 71–91, closing = 91–95. | |


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## Learning Outcomes

By the end of this course, learners will be able to:

1. Explain the agentic loop and apply context-engineering techniques in Claude Code.
2. Create custom slash commands, connect MCP tools and apply agent skills to extend Claude Code.
3. Use sub agents and hooks to automate specialised tasks, and apply AI security controls.
4. Build, deploy, test, secure and optimise a real website end-to-end with Claude Code.

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## Daily Schedule (9:00 AM – 6:00 PM, one-day class)

This is a **one-day class**, so the **1-hour assessment runs from 5:00 – 6:00 PM** and all teaching content finishes by 5:00 PM. Breaks are shortened (10-minute breaks, 45-minute lunch) to make room.

| Time | Duration | Topic / Activity | Method | Slides |
|---|---|---|---|---|
| 9:00 – 9:30 | 30 min | **Welcome & Administration** — Digital attendance (AM), trainer introduction, learner introductions, ground rules, course outline | Lecture / Discussion | 1–13 |
| 9:30 – 10:30 | 60 min | **Topic 1.1 Claude Code fundamentals — What makes an AI agent (model + harness + tokens; seven parts); how it works (agentic loop + built-in tools); Setup (Terminal, VS Code, Desktop, Cloud) & pricing; permission modes & bypass (VS Code and Desktop settings)** | Lecture / Demo | 14–30, 38–42 |
| 10:30 – 10:40 | 10 min | **Morning Break** | — | — |
| 10:40 – 11:20 | 40 min | **Topic 1.2 Context, memory & sessions** — context window & reading /context, CLAUDE.md vs auto memory, managing sessions, scheduled tasks, /goal | Lecture / Demo | 44–52 |
| 11:20 – 12:30 | 70 min | **Activity 1 — Build & Deploy a Website — Build a responsive software-dev site (hero, testimonials, enquiry form) after setting up the VS Code extension, or the Desktop app with a saved environment, create a GitHub repository and push with Claude, deploy on GitHub Pages via Actions, add** | Hands-on | 31–37, 43, 53–56 |


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| | | **README (evidence for PP Task 1)** | | |
| 12:30 – 1:15 | 45 min | **Lunch Break** | — | — |
| 1:15 – 1:45 | 30 min | **Topic 2.1 Custom Commands** — Slash commands, .claude/commands, arguments; Activity 2A — Create /gitpush command | Lecture / Hands-on | 57–61 |
| 1:45 – 2:25 | 40 min | **Topic 2.2 MCP Tools** — MCP architecture, connecting servers & scopes, Playwright MCP; Activity 2B — screenshot site & push; Activity 2C — test enquiry form to email | Lecture / Hands-on | 62–66 |
| 2:25 – 3:05 | 40 min | **Topic 2.3 Agent Skills** — SKILL.md and its front matter (name, description), project skills, installing skills; Activity 2D — install frontend-design & seo-audit skills and redesign the site (evidence for PP Task 2) | Lecture / Hands-on | 67–70 |
| 3:05 – 3:15 | 10 min | **Afternoon Break** | — | — |
| 3:15 – 3:45 | 30 min | **Topic 3.1 Sub Agents** — Creating sub agents and their front matter (name, description, tools, skills, model), sub agents vs agent teams; Activity 3A — Security sub agent + lead-magnet UI/UX sub agent | Lecture / Hands-on | 71–76 |
| 3:45 – 4:20 | 35 min | **Topic 3.2 Hooks** — Hook events; PreToolUse and PostToolUse use cases (validation; processing a refund on the tool outcome); a RAG chatbot that validates the refund request with PreToolUse and processes it with PostToolUse; demos (10-second pop-up, voice message on chatbot click); Activity 3B — build your own hooks (evidence for PP Task 3) | Lecture / Hands-on | 77–83 |
| 4:20 – 4:50 | 30 min | **Topic 3.3 AI Security** — Data classification, authorisation (read, write, edit, send), human gating and escalation, logging and a data-leak case, right-sized models; Activity 3C — test, secure | Lecture / Hands-on | 84–90 |


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| | | & deploy | | |
| 4:50 – 5:00 | 10 min | **Summary, Q&A & Closing** — Key-concepts recap (what triggers what), recap, digital attendance, TRAQOM survey & certification | Discussion | 91–95 |
| 5:00 – 5:30 | 30 min | **Written Assessment (WA)** — open book, on the LMS | Assessment | — |
| 5:30 – 6:00 | 30 min | **Practical Performance (PP)** — open book, on the LMS | Assessment | — |

*The 1-hour assessment (30 min WA + 30 min PP) runs from 5:00 PM and the class ends at 6:00 PM. Breaks are shortened (10-minute breaks, 45-minute lunch) to fit the assessment. Courseware and the assessment are available on the LMS at https://lms-tms.tertiaryinfotech.com/. Timings may be adjusted slightly to suit class pace.*

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## Topic-by-Topic Breakdown

### Topic 1 — Claude Code Fundamentals (≈ 1 hr 15 min + Activity 1) · Slides 14–56

- What makes an AI agent: model + harness + tokens; the seven parts (model, tools, skills, memory, knowledge base, instructions, runtime controls) and where each lives in Claude Code
- How Claude Code works: the agentic loop and built-in tools
- Setup: Terminal, IDE (VS Code), the Desktop app and the cloud (claude.ai/code); pricing and checking plan limits with /usage
- Context engineering & the context window (/context and how to read its token breakdown, /compact, /clear)
- Memory files: CLAUDE.md vs auto memory (/init, /memory)
- Permission modes, including the Claude Desktop mode menu (Auto, Manual, Accept edits, Plan, Bypass permissions), and the dangers of bypass permissions; allowing bypass mode in VS Code and Claude Desktop (Settings → Code)
- Managing sessions, scheduled tasks (Routines), and /goal
- **Activity 1: Build and deploy a website — set up the VS Code extension (secondary side bar) or the Desktop app with a saved environment, try a cloud session at claude.ai/code, then create a GitHub repository and push with Claude (Step 5)**

### Topic 2 — Commands, Tools and Skills (≈ 1 hr 50 min incl. Activity 2) · Slides 57–70

- Custom slash commands and arguments
- MCP tools and architecture; Playwright MCP
- Agent skills and installation; inside a SKILL.md — the name and description front matter decide when a skill fires; SKILL.md vs CLAUDE.md
- Activity 2 (Lab 2): 2A Create a /gitpush custom command; 2B Install Playwright MCP, screenshot the site, push; 2C Test the enquiry form and confirm enquiries reach your email; 2D Install & run frontend-design and seo-audit skills and redesign the site
- PP Task 2 evidence: a screenshot of the redesigned site from Activity 2D


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### Topic 3 — Agents, Hooks and Security (≈ 1 hr 35 min incl. Activity 3) · Slides 71–91

- Sub agents — front matter: name, description, tools, skills and model — and agent teams
- Hooks; PreToolUse use cases (validate or block) and PostToolUse use cases (act on the tool outcome, such as processing a refund); a RAG refund chatbot guarded by both hooks; demos — a 10-second pop-up and a voice message on chatbot click
- AI security — one slide per principle: data control (classification), permission control (authorisation: read, write, edit, send), human in the loop (gating and escalation), traceability (logging and a data-leak case) and right-sized models (the more powerful the model, the harder to control)
- Activity 3 (Lab 3): 3A Security and UI/UX (lead-magnet) sub agents; 3B Build your own hooks — a 10-second pop-up and a voice message on chatbot click; 3C Test, secure and deploy
- PP Task 3 evidence: a screenshot of a hook the learner built (any hook)
- Key concepts summary: what triggers what — CLAUDE.md (every session), skills and tools (on demand), slash commands (manual), hooks (by event) and sub agents (by delegation)

## Lab Materials

- Lab 1 (Topic 1, Activity 1, PP Task 1) — labs/lab1-seven-steps/: the 7-step build workflow, one file per step, plus a reference site and three sample sites.
- Lab 2 (Topic 2, Activity 2, PP Task 2) — labs/lab2-commands-tools-skills/: a /gitpush custom command, Playwright MCP screenshots in the README, and installing skills to harden and redesign the Lab 1 site.
- Lab 3 (Topic 3, Activity 3, PP Task 3) — labs/lab3-agents-hooks-security/: a security-scan sub agent, a pop-up hook, a voice hook, a Claude Code lifecycle hook, and the final test, secure and deploy. An add-task sub agent and /loop are take-home parts.
- Labs repository: https://github.com/tertiarycourses/TGS-2025052468-Agentic-AI-Applications-with-Claude-Code/tree/main/labs

## Resources Required

- Computers with internet access (per learner)
- Claude Code installed and signed in
- GitHub accounts; Node.js installed
- Projector / shared screen for demonstrations
- Course slides and this learner guide

## Assessment

- **Written Assessment (WA):** 30 min — open book, on the LMS
- **Practical Performance (PP): 30 min — open book, on the LMS; three screenshot tasks, one per topic and activity**
- Assessment starts at 5:00 PM and the class ends at 6:00 PM.
- **Courseware & assessment:** available on the LMS at https://lms-tms.tertiaryinfotech.com/
- **Funding criteria:** minimum 75% attendance and assessed as *Competent*


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